PROBLEM 22

Particle on a Curve

Original statement · Problems.pdf, p. 10

1D Curve Constraint

A heavy point particle P of mass m is constrained to move without friction along the curve:

$$y = a\left(1 - \cos\frac{x}{a}\right) \tag{4}$$

in the range $-\pi a \leq x \leq \pi a$, where a is a positive constant. Initially, the particle is located at $x = 0$ with velocity v_0 along the positive x axis.

(a) Reduce the problem to integrals.

(b) If $v_0 = \sqrt{ga}$, determine the minimum and maximum value of x along the motion.

(c) What should be the minimum value of v_0 in order to reach $x = \pi a$?

Extended Section (Moving Reference Frame):
Let's assume now that the curve containing the point particle rotates around the OY axis with constant angular velocity $\boldsymbol{\omega} = \Omega\mathbf{j}$.

(d) Define a convenient moving reference frame with origin in O, $OX_1Y_1Z_1$. Determine the inertia force that acts on the particle in this moving reference frame as a function of m, Ω, x_1 and $\dot{x}_1$, where x_1 represents the position coordinate of P along OX_1.

(e) Reduce the problem to quadratures and determine the value of the reaction force.

(f) Find out a transcendental equation whose solutions are the equilibrium positions of the point particle in $OX_1Y_1Z_1$.

(g) If Ω were a function of time, would it be possible to use the energy equation in the same way as before? Justify your answer.

Full Analytical Solution (4-Phase Methodology)

PHASE 1

Physical Statement, Hypotheses & Parameters

- **Physical System:** A particle of mass m is constrained to move without friction along the smooth planar curve:

$$y = a\left(1 - \cos\frac{x}{a}\right), \qquad x \in [-\pi a, \pi a]$$

in a vertical plane under uniform downward gravity $\mathbf{g} = -g\mathbf{j}$.

- **CDOFs:** 1 degree of freedom, parametrized by Cartesian abscissa $x(t)$.
- **Initial State ($t = 0$):** $x(0) = 0 \implies y(0) = a(1 - \cos 0) = 0$, with initial velocity $\mathbf{v}_0 = v_0\mathbf{i}$ ($v_0 > 0$).
- **Parameters:** $m > 0$ [kg], $a > 0$ [m], $g > 0$ [m/s^2], $v_0 > 0$ [m/s].

PHASE 2

Reference Frames, Vector Bases & Coordinate Geometry

- **Inertial Frame:** $\mathcal{F}_0 = \{O; \mathbf{i}, \mathbf{j}, \mathbf{k}\}$.
- **Curve Differential Elements:**

$$\frac{dy}{dx} = \frac{d}{dx}\left[a\left(1 - \cos\frac{x}{a}\right)\right] = \sin\left(\frac{x}{a}\right)$$

Differential arc-length:

$$ds^2 = dx^2 + dy^2 = dx^2 + \sin^2\left(\frac{x}{a}\right)dx^2 = \left[1 + \sin^2\left(\frac{x}{a}\right)\right]dx^2$$

$$ds = \sqrt{1 + \sin^2\left(\frac{x}{a}\right)}\,dx$$

- **Velocity:**

$$v = \frac{ds}{dt} = \sqrt{1 + \sin^2\left(\frac{x}{a}\right)}\,\dot{x}$$

PHASE 3

Step-by-Step Mathematical Deduction

Part (a): Reduction of Motion to Integrals

Pedagogical Justification: The frictionless constraint does zero work ($W_N = 0$). Gravity is conservative with potential energy $V(y) = mgy = mga(1 - \cos(x/a))$. Total mechanical energy is constant.

$$E = \frac{1}{2}mv^2 + mgy = \frac{1}{2}mv_0^2 + mg(0) = \frac{1}{2}mv_0^2$$

$$\frac{1}{2}m\left[1 + \sin^2\left(\frac{x}{a}\right)\right]\dot{x}^2 + mga\left(1 - \cos\frac{x}{a}\right) = \frac{1}{2}mv_0^2$$

Multiply by $2/m$:

$$\left[1 + \sin^2\left(\frac{x}{a}\right)\right]\dot{x}^2 = v_0^2 - 2ga\left(1 - \cos\frac{x}{a}\right)$$

Isolate $\dot{x} = \frac{dx}{dt}$:

$$\frac{dx}{dt} = \pm\sqrt{\frac{v_0^2 - 2ga(1 - \cos(x/a))}{1 + \sin^2(x/a)}}$$

Separation of variables yields the quadrature:

$$\int_0^{x(t)} \sqrt{\frac{1 + \sin^2(x'/a)}{v_0^2 - 2ga(1 - \cos(x'/a))}}\, dx' = \pm t$$

Part (b): Turning Points for $v_0 = \sqrt{ga}$

> **Pedagogical Justification:** Turning points occur where the velocity vanishes ($v = 0 \iff \dot{x} = 0$).

$$v_0^2 - 2ga\left(1 - \cos\frac{x}{a}\right) = 0$$

Substitute $v_0^2 = ga$:

$$ga - 2ga\left(1 - \cos\frac{x}{a}\right) = 0$$

Divide by ga:

$$1 - 2\left(1 - \cos\frac{x}{a}\right) = 0 \implies 2\cos\frac{x}{a} = 1 \implies \cos\frac{x}{a} = \frac{1}{2}$$

In the domain $x/a \in [-\pi, \pi]$:

$$\frac{x}{a} = \pm\frac{\pi}{3} \implies x_{min} = -\frac{\pi}{3}a, \qquad x_{max} = +\frac{\pi}{3}a$$

Part (c): Minimum Velocity v_0 to Reach $x = \pi a$

> **Pedagogical Justification:** At the crest $x = \pi a$, the height is $y(\pi a) = a(1 - \cos\pi) = a(1 - (-1)) = 2a$. For the particle to reach this crest, its kinetic energy at the peak must be non-negative ($T(\pi a) \geq 0$).

$$E \geq V(\pi a) \implies \frac{1}{2}mv_0^2 \geq mg(2a) = 2mga$$

$$v_0^2 \geq 4ga \implies v_{0,min} = \sqrt{4ga} = 2\sqrt{ga}$$

Part (d): Moving Reference Frame with Rotation $\boldsymbol{\omega} = \Omega\mathbf{j}$

- Frame $\mathcal{F}_1 = \{O; \mathbf{i}_1, \mathbf{j}, \mathbf{k}_1\}$ rotating with $\boldsymbol{\Omega} = \Omega\mathbf{j}$.
- Particle position: $\mathbf{r}' = x_1\mathbf{i}_1 + y(x_1)\mathbf{j}$.
- Relative velocity: $\mathbf{v}' = \dot{x}_1\mathbf{i}_1 + \sin(x_1/a)\dot{x}_1\mathbf{j}$.
- Inertia forces in $\mathcal{F}_1$:

1. Centrifugal Force:

$$\mathbf{F}_{cf} = -m\boldsymbol{\Omega} \times (\boldsymbol{\Omega} \times \mathbf{r}') = m\Omega^2 x_1 \mathbf{i}_1$$

2. Coriolis Force:

$$\mathbf{F}_{Cor} = -2m\boldsymbol{\Omega} \times \mathbf{v}' = -2m(\Omega \mathbf{j}) \times (\dot{x}_1 \mathbf{i}_1 + \dot{y}_1 \mathbf{j}) = 2m\Omega \dot{x}_1 \, \mathbf{k}_1$$

Part (e): Quadratures and Reaction Force in the Moving Frame

- Effective potential in $\mathcal{F}_1$:

$$V_{eff}(x_1) = mgy(x_1) - \frac{1}{2}m\Omega^2 x_1^2 = mga\left(1 - \cos\frac{x_1}{a}\right) - \frac{1}{2}m\Omega^2 x_1^2$$

- Energy conservation in $\mathcal{F}_1$:

$$E' = \frac{1}{2}m\left[1 + \sin^2(x_1/a)\right]\dot{x}_1^2 + mga\left(1 - \cos\frac{x_1}{a}\right) - \frac{1}{2}m\Omega^2 x_1^2 = \text{constant}$$

- Reaction force: $\mathbf{N} = N_n \mathbf{n} + N_z \mathbf{k}_1$.

Along the binormal $\mathbf{k}_1$: $N_z = -F_{Cor,z} = -2m\Omega\dot{x}_1$.

Part (f): Transcendental Equilibrium Equation

Equilibrium occurs where $\frac{dV_{eff}}{dx_1} = 0$:

$$\frac{dV_{eff}}{dx_1} = mg\sin\left(\frac{x_1}{a}\right) - m\Omega^2 x_1 = 0 \implies g\sin\left(\frac{x_1}{a}\right) - \Omega^2 x_1 = 0$$

Part (g): Case $\Omega = \Omega(t)$

If Ω varies with time, the centrifugal potential becomes explicitly time-dependent ($\frac{\partial V_{cf}}{\partial t} = -m\Omega\dot{\Omega}x_1^2 \neq 0$) and an Euler inertia force $\mathbf{F}_{Euler} = -m\dot{\mathbf{\Omega}} \times \mathbf{r}'$ appears, which does non-zero work. Thus, mechanical energy in $\mathcal{F}_1$ is **not** conserved.

PHASE 4

Physical Interpretation & Dimensional Verification

- The potential well has width $2\pi a$ and barrier height $2ga$. Rotation creates a centrifugal destabilizing potential $-\frac{1}{2}m\Omega^2 x_1^2$ so that the origin loses stability for $\Omega^2 > g/a$. Off-centre equilibria satisfy $\sin u = \lambda u$ with $u = x_1/a$ and $\lambda = \Omega^2 a/g$; since $\sin u/u$ decreases monotonically from 1 to 0 on $(0, \pi]$, they exist (one pair $\pm u^*$ with $0 < u^* < \pi$) only for $\lambda < 1$, i.e. $\Omega^2 < g/a$, and there they are unstable ($V''_{eff} = (mg/a)(\cos u^* - \lambda) < 0$).

> **Erratum:** The first version stated that off-centre equilibria appear for $\Omega^2 > g/a$; they exist only for $\Omega^2 < g/a$ (subcritical case) and are unstable. Numeric check: a scan of $\sin u = \lambda u$ on $(0, \pi]$ finds one root for $\lambda = 0.5$ ($u^* = 1.8955$) and for 0.9, and none for $\lambda = 1, 1.1, 2$.

Source: Problems.pdf (Page 11)